

Parking Management System

```
#include <iostream>
#include <fstream>
#include <ctime>
using namespace std;

class Parking {
private:
    int num_auto=0, num_bus=0, num_car=0;
    int amount=0, count=0;
    int vehicleID[100];
    string vehicleType[100];
    int nextID;

public:
    // Constructor
    Parking() {
        loadFromFile();

        if (count == 0)
            nextID = 1;
        else if (count > 0)
            nextID = vehicleID[count - 1] + 1;
    }

    int menu() {
        int ch;
        cout << "\n If You Want to Park Bus Enter '1'";
        cout << "\n If You Want to Park Car Enter '2'";
        cout << "\n If You Want to Park Auto Enter '3'";
        cout << "\n If You Want to Show Status Enter '4'";
        cout << "\n If You Want to Leave Enter '5'";
        cout << "\n If You Want to Delete Data Enter '6'";
        cout << "\n If You Want to Exit From Program Enter '7'";
        cout << "\n Please Enter Your Choice : ";
        cin >> ch;
        return ch;
    }

    // Function to get current date & time
    void showTime() {
        time_t now = time(0);
        char* dt = ctime(&now);
    }
}
```

```
    cout << " Date & Time: " << dt;
}
```

```
void saveToFile() {
    ofstream file("/Users/pruthvipatel/Desktop/parking_data3.txt");
    file << num_bus << " " << num_car << " " << num_auto << " "
    << amount << " " << count << endl;
}
```

```
    for (int i = 0; i < count; i++) {
        file << vehicleID[i] << " " << vehicleType[i] << endl;
    }
    file.close();
}
```

```
void loadFromFile() {
    ifstream file("/Users/pruthvipatel/Desktop/parking_data3.txt");

    if (file) {
        file >> num_bus >> num_car >> num_auto >> amount >> count;

        for (int i = 0; i < count; i++) {
            file >> vehicleID[i] >> vehicleType[i];
        }
    } else {
        num_auto = num_bus = num_car = 0;
        amount = 0;
        count = 0;
        nextID=1;
    }

    file.close();
}
```

```
void parkVehicle(string type, int charge)
{
    int id = nextID++;

    vehicleID[count] = id;
    vehicleType[count] = type;
    cout << id << endl;

    saveReceipt(type,id,charge);
}
```

```

void saveReceipt(string type, int id, int charge) {
    ofstream file("/Users/pruthvipatel/Desktop/receipt3.txt", ios::app);

    file << "\n----- OP Parking Service -----\\n";
    file << "Vehicle Type: " << type << endl;
    file << "Unique ID: " << id << endl;
    file << "Charge: " << charge << endl;
    file << "Time: ";
    time_t now = time(0);
    file << ctime(&now);
    file << "Thank you for choosing us!\\n";
    file << "-----\\n";

    file.close();
}

```

```

void bus() {
    cout << " BUS PARKED\\n";
    cout << " VEHICLE'S ID : ";
    parkVehicle("BUS", 300);
    showTime();
    num_bus++;
    amount += 300;
    count++;
    saveToFile();
}

```

```

void car() {
    cout << " CAR PARKED\\n";
    cout << " VEHICLE'S ID : ";
    parkVehicle("CAR", 150);
    showTime();
    num_car++;
    amount += 150;
    count++;
    saveToFile();
}

```

```

void autoRickshaw() {
    cout << " AUTO RICKSHAW PARKED\\n";
    cout << " VEHICLE'S ID : ";
    parkVehicle("AUTO", 100);
    showTime();
    num_auto++;
    amount += 100;
    count++;
    saveToFile();
}

```

```

void showDetail() {
    cout << "\nCurrent Status of OP Parking Service\n";
    cout << "Number of bus = " << num_bus << endl;
    cout << "Number of car = " << num_car << endl;
    cout << "Number of auto rickshaw = " << num_auto << endl;
    cout << "Total Vehicles = " << count << endl;
    cout << "Total Gain Amount = " << amount << endl;
}

```

```

void exitV()
{
    int ID;
    cout << "Enter Vehicle Id : ";
    cin >> ID;

    int pos = -1;

    for (int i = 0; i < count; i++)
    {
        if (vehicleID[i] == ID)
        {
            pos = i;
            break;
        }
    }

    if (pos != -1)
    {

        if (vehicleType[pos] == "BUS")
            num_bus--;
        else if (vehicleType[pos] == "CAR")
            num_car--;
        else if (vehicleType[pos] == "AUTO")
            num_auto--;

        for (int i = pos; i < count - 1; i++)
        {
            vehicleID[i] = vehicleID[i + 1];
            vehicleType[i] = vehicleType[i + 1];
        }

        count--;
    }
}

```

```

        cout << "Vehicle Exit Successful\n";
        cout << "Time: ";
        showTime();
    }
    else
    {
        cout << "❌ Invalid ID! Vehicle not found.\n";
    }

    saveToFile();
}

```

```

void deleteData() {
    num_bus = num_car = num_auto = 0;
    amount = 0;
    count = 0;
    nextID=1;
    saveToFile();
    cout << "All data cleared!\n";
}
};

```

```

int main() {
    Parking p;
    int choice;

    while (true) {
        choice = p.menu();

        switch (choice) {
            case 1: p.bus(); break;
            case 2: p.car(); break;
            case 3: p.autoRickshaw(); break;
            case 4: p.showDetail(); break;
            case 5: p.exitV(); break;
            case 6: p.deleteData(); break;
            case 7: return 0;
            default: cout << "Invalid Choice\n";
        }
    }

    return 0;
}

```